Project-Specific Robustness and Ablation Analysis

1. Objective

The objective of this analysis was to identify weaknesses in underwater object detection and understand how class imbalance, object characteristics and training strategies influenced model performance.

The analysis used the existing validation results from the V2.1 YOLO11s baseline and the completed oversampling experiment.

2. Class-wise performance analysis

The V2.1 YOLO11s baseline achieved an aggregate mAP@50 of 0.236.

Class	**Validation mAP@50**
rov	0.574
plant	0.000197
animal_fish	0.468
animal_starfish	0
animal_shells	0
animal_crab	0.742
animal_eel	0.480
animal_etc	0.0075
trash_etc	0.184
trash_fabric	0.448
trash_fishing_gear	0
trash_metal	0.478
trash_paper	0.0000353
trash_plastic	0.385
trash_rubber	0
trash_wood	0.00542

The model performed better on classes such as animal_crab, rov, animal_fish, animal_eel and trash_metal.

However, several classes had very low or zero mAP. This indicates that aggregate metrics alone do not represent performance across all object categories.

3. Class imbalance investigation

The V2.1 dataset contained considerable differences in class frequencies.

Examples:

- rov: 3,316 instances
- trash_etc: 2,039 instances
- trash_plastic: 1,864 instances
- trash_rubber: 142 instances
- trash_paper: 193 instances
- trash_fishing_gear: 196 instances

Rare categories had fewer training examples, making generalization more difficult.

4. Oversampling ablation

A targeted oversampling strategy was applied to the V2.1 training data. Images containing underrepresented classes were repeated according to predefined class-specific factors.

The validation and test sets were left unchanged.

Metric	**YOLO11s baseline**	**YOLO11s oversampled**
Precision	0.519	0.494
Recall	0.234	0.198
mAP@50	0.236	0.203
mAP@50–95	0.143	0.125

The oversampling experiment did not improve overall performance.

Some individual classes improved, including animal_eel (0.480 to 0.594 mAP@50) and trash_fabric (0.448 to 0.506). However, the aggregate metrics decreased.

A possible explanation is that repeating images increases their frequency without introducing new visual information. In addition, image-level oversampling can repeat other objects present in the same image.

5. Robustness considerations

The following conditions are important for underwater detection:

Condition	Potential challenge	Current evidence
Low visibility	Reduced contrast and object visibility	Not separately stress-tested
Small objects	Limited visual information and fewer pixels	Class-wise results indicate uneven detection performance; no dedicated size-stratified evaluation
Occlusion	Objects partially hidden by marine vegetation or other objects	Not separately stress-tested
Clutter	Multiple overlapping marine objects	Not separately stress-tested
Class imbalance	Rare categories receive fewer training examples	Supported by class-frequency and oversampling analysis

6. Research insights

1. A high aggregate mAP does not guarantee good performance for every class.
2. Class imbalance is a significant challenge in the TrashCAN dataset.
3. Repeating rare-class images does not necessarily improve generalization.
4. Dataset quality and leakage-safe splitting are important for reliable evaluation.
5. Further experiments should focus on class-aware sampling, targeted augmentation and class-wise error analysis.

7. Limitations and future work

This analysis is based on existing validation metrics and completed training experiments. Dedicated low-visibility, occlusion, clutter and object-size stress tests were not completed as part of the recorded experiments.

Future work should include controlled image degradation, object-size-specific evaluation, confidence-threshold analysis and more diverse training samples for underrepresented categories.

Conclusion

The analysis highlighted class imbalance and uneven class-wise detection performance as important limitations of the current model. The oversampling experiment did not improve aggregate metrics, indicating that more effective strategies than simple image repetition are needed.